



UNIVERSITY OF
BIRMINGHAM

Forensic

v

Computational Authorship Analysis

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Today

- Session 1.1 – 14.00-15.30
 - Intro to Forensic Linguistics
 - Intro to Authorship Analysis
- Session 1.2 – 15.45-17.00
 - Practical Application

Bitcoin

- The founder of Bitcoin is a person named Satoshi Nakamoto
- However, it remains a mystery who Nakamoto actually is as the person he claimed to be does not exist
- There are many theories as to why the founder chooses or chose to remain anonymous or working under a false name



Bitcoin

1997: Adam Back proposes hashcash

1998: Wei Dai proposes b-money and
Nick Szabo proposes bit-gold

2007: Hal Finney proposes reusable
proofs of work

2008-8-18: bitcoin.org registered

2008-10-31: Satoshi Nakamoto posts 'Bitcoin:
A Peer-to-Peer Electronic Cash System'

2009-1-3: Nakamoto mines first bitcoin block

2009-1-12: Finney receives first bitcoin
transaction from Nakamoto

2009-11-22: Bitcoin Talk Forum goes online

2010-12-12: Last post from Nakamoto on
forum

2012-9: Gavin Andresen becomes chief
scientist for newly founded Bitcoin Foundation

Bitcoin

The Bitcoin Whitepaper

- Published on October 31, 2008
- Title: *Bitcoin: A Peer-to-Peer Electronic Cash System*
- Introduced the concept of decentralised digital currency
- Eliminates intermediaries (banks) through blockchain technology
- Lists Satoshi Nakamoto as the author / originator of Bitcoin
 - Anonymous person or group who wrote the whitepaper
 - Active in forums & emails until 2011, then disappeared
 - Holds over 1 million BTC (~\$60B+ today)

Bitcoin

Many possible candidates, for example:

- Hal Finney (Cryptographer & first Bitcoin transaction recipient)
- Nick Szabo (Created “Bit Gold,” similar to Bitcoin)
- Dorian Nakamoto (Journalist mistake, but denied being Satoshi)
- Craig Wright (Claims to be Satoshi but lacks proof)
- Wei Dai (Creator of “b-money,” a precursor to Bitcoin)
- Martti Malmi (Early contributor to Bitcoin development)
- ...

Bitcoin

- The Bitcoin Talk Forum, which has been running since late 2009, is an especially rich dataset for investigating Satoshi's identity (at least on the forum!)
- There are a substantial numbers of posts by Satoshi, as well as major candidates, various minor candidates, and many anonymous and pseudo-anonymous users

The screenshot shows the Bitcoin Forum website. At the top, the header reads "Bitcoin Forum" with a logo and the tagline "simple machines forum". Below the header, the date and time "June 12, 2025, 11:42:31 AM" are displayed. A welcome message for a guest user is shown, along with a news banner for the latest Bitcoin Core release (29.0) and a search bar. A navigation bar includes links for HOME, HELP, SEARCH, LOGIN, REGISTER, and MORE.

The main content area is titled "Bitcoin Forum" and features a table of board listings. The first board is "Bitcoin", which includes a description of general discussion about the Bitcoin ecosystem, a count of 271,795 posts and 10,288 topics, and the last post by Wind_FURY. The second board is "Development & Technical Discussion", which includes a description of technical discussion about Satoshi's Bitcoin client and the Bitcoin network, a count of 34,910 posts and 2,637 topics, and the last post by Jakow Don.

Bitcoin		
 Bitcoin Discussion General discussion about the Bitcoin ecosystem that doesn't fit better elsewhere. News, the Bitcoin community, innovations, the general environment, etc. Discussion of specific Bitcoin-related services usually belongs in other sections. <i>Moderator: hilariousandco</i>	271795 Posts 10288 Topics	Last post by Wind_FURY in <i>Re: 2 Euros worth of Bit...</i> on Today at 11:39:32 AM
Child Boards: Legal , Press , Meetups , Important Announcements		
 Development & Technical Discussion Technical discussion about Satoshi's Bitcoin client and the Bitcoin network in general. No third-party sites/clients, bug reports that do not require much discussion (use github), or support requests. <i>Moderators: gmaxwell, achow101</i>	34910 Posts 26370 Topics	Last post by Jakow Don in <i>What is the value of tha...</i> on Today at 11:37:21 AM

Jaccard Distance

- A measure of dissimilarity between two sets, based on their shared and unique elements
- Helps compare sets in various applications like text analysis, recommendation systems, and clustering
- Works well with binary and categorical data

Jaccard Distance

- You can measure the distance between two texts (or between every two texts in a collection of texts)
- Do to this we divide the number of word types that occur in both texts (overlap) by the number of word types across the texts
- If two texts have zero words in common, the distance is 1; if they have all words in common, the distance is 0

Jaccard Distance

$(A \cap B)$

Area of Overlap



$$J(A, B) = \frac{(A \cap B)}{(A \cup B)}$$

$$Jdist(A, B) = 1 - J(A, B)$$

Area of Union

$(A \cup B)$



Jaccard Distance

Text 1: the dog chased **the** cat down **the** street [6]

Text 2: everyone walked out of the building carefully **of** course [8]

Text 3: a bit further down the street, everyone has **a** bigger house [10]

types



Jaccard Distance



Jaccard Distance



Jaccard Distance



$$J(A, B) = (A \cap B) / (A \cup B)$$

$$Jdist(A, B) = 1 - J(A, B)$$

$$1 - [1 / (5 + 1 + 7)] = .92$$

Jaccard Distance

$$1 - [3 / (3+3+8)] = .76$$

Text 1

Text 3

3

2

7

1

1

6

Text 2

$$1 - [1 / (5+1+7)] = .92$$

$$1 - [2 / (9+2+6)] = .88$$

Jaccard Distance

Text 1: the dog chased **the** cat down **the** street [6]

Text 2: everyone walked out of the building carefully **of** course [8]

Text 3: a bit further down the street, everyone has **a** bigger house [10]

	Text 1	Text 2	Text
Text 1	0		
Text 2	0.92	0	
Text 3	0.76	0.88	0

*Text 1 and Text 3 are
the most similar*

Practical Application

Images

- https://upload.wikimedia.org/wikipedia/commons/c/ce/Bust_of_Satoshi_Nakamoto_in_Budapest.jpg